INCH-POUND
MIL-C-17361/11A(SH)
13 March 1995
SUPERSEDING
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10 March 1989

MILITARY SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPE AQB - LF400 AND AQB - LF402

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-17361.

AMSC N/A
FSC 5925

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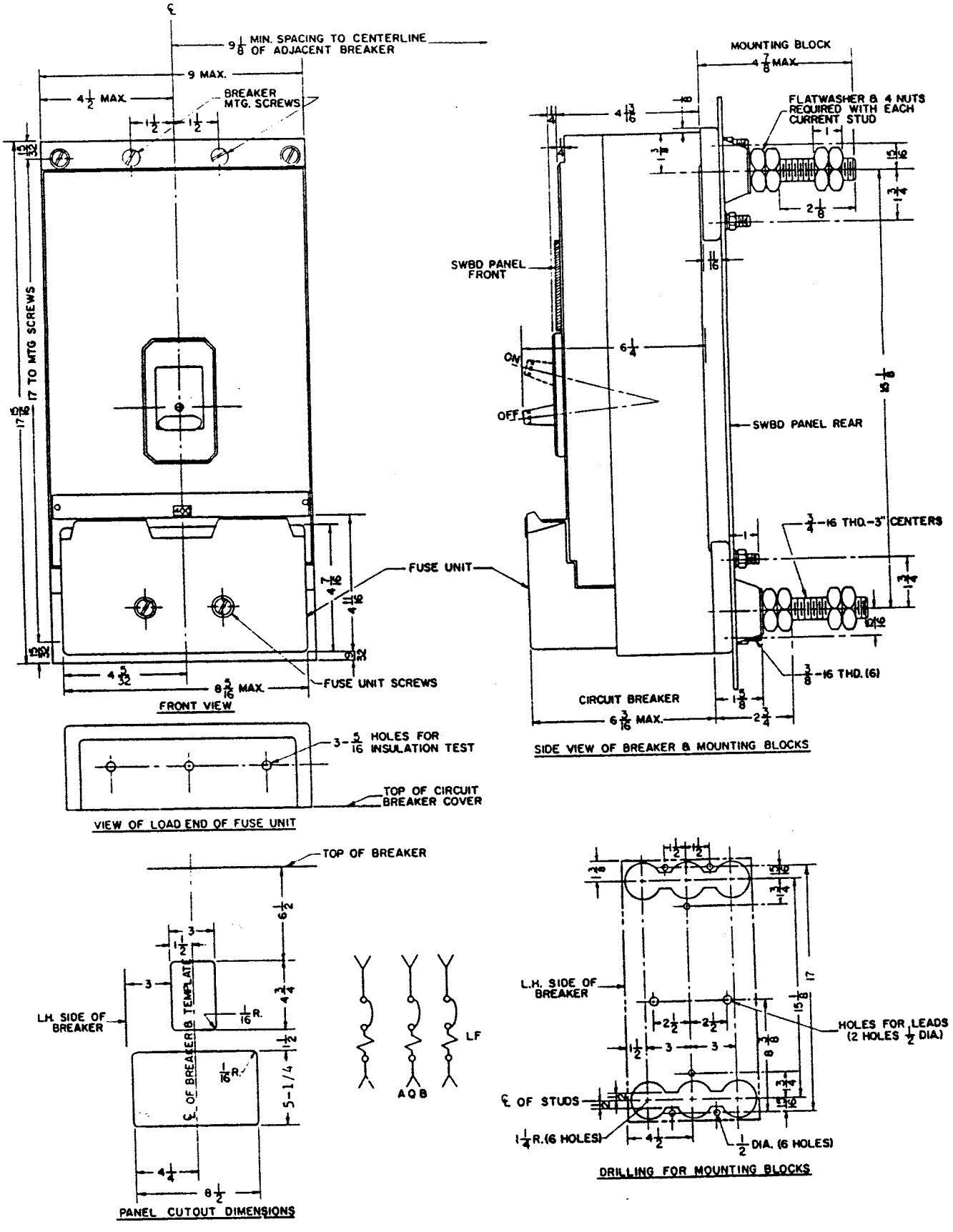


FIGURE 1. Circuit Breaker Type AQB-LF400 and AQB-LF402 outline and mounting dimensions

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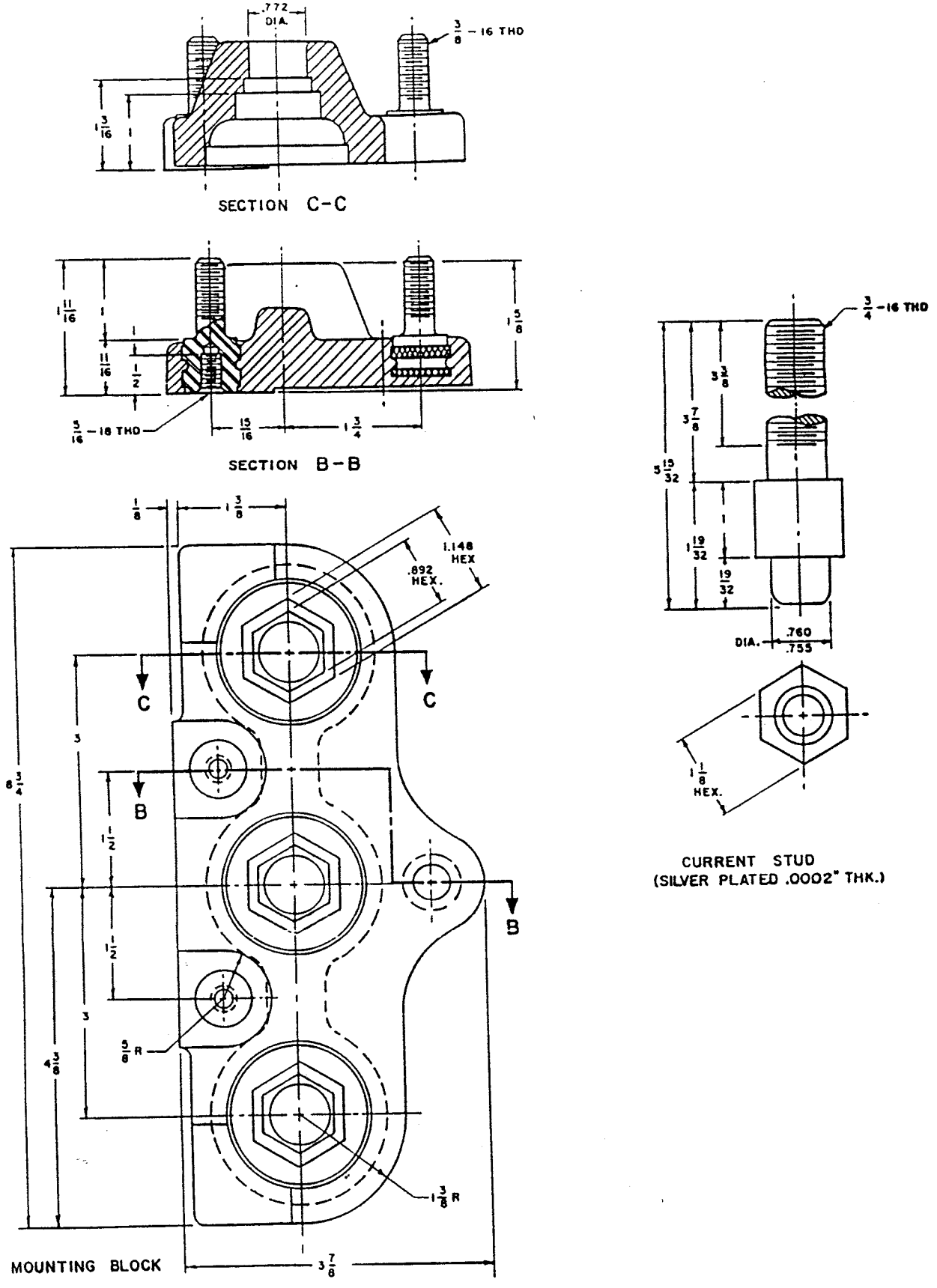


FIGURE 2. Mounting block for circuit breaker type AQB-LF400 and AQB-LF402.

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REQUIREMENTS:

Dimensions: See figures 1 and 2.
Maximum rated continuous current: 400 amperes.
Voltage and frequency: 500 Vac, 60 Hz.
250 Vdc.
Interrupting rating: 100,000 amperes (60 Hz ac).
Peak let-through current: 45,000 amperes.
Trip unit: Interchangeable.

Protection (for trip units type AQB-LF400 only):
Trip unit rating (amperes): See table I.
LTD and adjustable instantaneous, all 3 poles.
LTD calibration: See table II.
Instantaneous calibration: See table I.

Protection (for trip units type AQB-LF402 only):
Frame rating: 400 amperes (60 Hz ac only, no dc rating).
Continuous current settings (CCS): 250, 300, 350, and 400 amperes.
LTD, STD, and Instantaneous, all 3 poles.
LTD calibration: See table IIa.
STD calibration: The short time delay pickup settings shall be 2X, 3X, 4X, 6X, 8X, 10X, and 15X the continuous current settings. The short time delay "delay settings" shall be as specified in Table IIb.
Instantaneous calibration: The instantaneous pickup settings shall be 3X, 4X, 6X, 8X, 10X, 12X, 15X, 20X, 30X, and 40X the frame rating. Production units shall be calibrated at the 8X setting. Opening time shall be 0.025 seconds or less when tested at 3X the instantaneous pickup (for qualification tests only, not for routine calibration).
Time characteristics curves (TCC): TCC shall be developed and provided. Test points listed for LTD, STD, and Instantaneous shall comply with the TCC.

Current limiting fuse: LF5H250RD, 14,000 - 20,000 amperes.
LF5H400RD, 17,000 - 30,000 amperes.
Type of connections: Back (removable mtg.).
Electric operator: Available.
Undervoltage trip or shunt trip: Either available.
Auxiliary switch: Two (a) contacts; two (b) contacts.
Handle boot: Not available.
Handle lock: Group 3. (Figure 1 of MIL-C-17361)
Electrical test: The rate of operation shall be 5 operations per minute for a total of 4000 close-open operations.
Temperature rise tests: See table III.
Interrupting tests: See table IV.

NOTE: NQB circuit breakers shall consist of an AQB type without trip unit and a modification kit of metallic parts.

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TABLE I. Adjustable instantaneous trip settings amperes/maximum time. 1/

Element rating 2/	LO 4/		2		3		4		HI		Seconds 3/	
	Ac	Dc	Ac	Dc	Ac	Dc	Ac	Dc	Ac	Dc	60 Hz ac	Dc
250T	880	---	1035	---	1165	---	1325	---	1530	---		
250TM	3190	---	3725	---	4325	---	5175	---	5825	---		
300T	1380	---	1610	---	1820	---	2035	---	2280	---		
300TM	3190	---	3725	---	4325	---	5175	---	5825	---	0.024	0.028
350T	1380	---	1610	---	1820	---	2035	---	2280	---		
350TM	3190	---	3725	---	4325	---	5175	---	5825	---		
400T	1380	---	1610	---	1820	---	2035	---	2280	---		
400TM	3190	---	3725	---	4325	---	5175	---	5825	---		

1/ The tolerance for each ampere setting shall not exceed plus or minus 10 percent.
2/ Identification marking of trip units:
Numerals indicate continuous current rating of the specific trip unit.
Letter "M" denotes calibration for applications having a high inrush current.
Letter "T" denotes that the trip units are available to AQB-LF400.
3/ Time shall be measured at three times pick-up current of instantaneous setting. For qualification tests only, not for routine calibration.
4/ The LO and HI settings only are calibrated. Others are approximate.

TABLE II. Tripping characteristics - long time delay (for trip units type AQB-LF400 only).

LTD trip element percent current	Tripping time 60 Hz ac trip element
130	Not less than 1 hour
225	Less than 1 hour
600	Between 20 and 32 seconds

TABLE IIa. Tripping characteristics - long time delay (for trip units type AQB-LF402 only).

LTD in Multiples of CCS	Tripping Time 60 Hz ac
1.35	Not less than 1 hour
1.65	Less than 1 hour
4.0	Between 35 and 55 seconds
6.0	Between 20 and 32 seconds

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TABLE IIb. Adjustable short-time delay "delay settings" in seconds (for trip units type AQB-LF402 only).

Short-Time Delay Pickup Setting	Short-Time Delay "Delay Setting"		
	1	2	3
All Settings	1/ .025 - .055	.065 - .095	.105 - .135

1/ Time delay with 2X short-time delay pickup settings shall be .025 to .065 seconds.

TABLE III. Size of cable and bus bars for temperature rise tests.

Copper rating-amperes	Cable size		Copper bus size (inch)
	Ac	Dc	Ac and Dc
400	Two AWG 000	400 MCM	1-1/2 x 1/8

TABLE IV. Interrupting tests for circuit breaker type AQB-LF400. 1/

Circuit sample	Element amperes	Interrupting circuit amperes	Power factor lagging	Unit operation	See note
1A	400	100,000	0.15-0.20	O-CO-CO	2/
1A(1)	400	17,000	0.15-0.20	O-CO-CO	3/ 4/
1A(1)		30,000	0.15-0.20	O-CO-CO	2/ 3/
1A(1)		75,000	0.15-0.20	O-CO-CO	2/ 3/
1B	250	100,000	0.15-0.20	O-CO-CO	2/

1/ Oscillograms shall be taken on all interrupting tests.
2/ Circuit breaker shall open and at least one fuse blow.
3/ New circuit breakers may be used for tests above the standard circuit breaker rating.
4/ Circuit breaker shall open on instantaneous trip and fuses shall not blow.

Preparing activity:
Navy - SH
(Project 5925-N170)

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